PAPER XXXVII · MARCH 2026 · hep-th / astro-ph.CO · PREPRINT — NOT PEER REVIEWED

Paper XXXVII

The Minimal Effective Lagrangian:

α as a Dynamical Invariant of a Constrained Field System

From Heuristic Subtraction to a Coupled Field Machine

Dervis Kadunic | Independent Researcher

Companion to Papers I–XXXVI | arXiv/Zenodo March 2026

ABSTRACT

Papers XXXII–XXXVI established the alpha formula $\alpha = (\varepsilon^2 - 3v)/\sqrt{15}$ and its conservation on the constraint surface. This paper upgrades the framework from a heuristic observation to a minimal effective Lagrangian. The central shift: α is no longer described as 'energy minus tax.' It is the stable invariant of a coupled field system $\{\phi, \varepsilon\}$ governed by a constraint potential $V_{constraint} = \lambda(F(\phi,\varepsilon) - \alpha_{■})^2$. We derive the Euler-Lagrange equations, show the restoring force explicitly, verify the stability condition $\partial^2 V_{eff}/\partial\varepsilon^2 > 0$ for all $\lambda > 0$, and present numerical simulations confirming that α settles to within 0.01 ppm of $\alpha_{■}$ within a few dynamical timescales for $\lambda \geq 1$. The single open parameter λ (stiffness) sets the settling rate. Deriving λ from the Schwarz P geometry is the target for Paper XXXVIII.

I. The Upgrade: From Accountant to Engineer

The progression across this series has been:

- **Papers XXXII–XXXIV (Accountant phase):** We found that $\alpha = (\varepsilon^2 - 3v)/\sqrt{15}$ matches the CODATA value to 99.01%. We derived each component from topology (Paper XXXIII) and bridged the units (Paper XXXIV).
- **Papers XXXV–XXXVI (Audit phase):** We stress-tested the framework and found that the subtraction language implies 42,000 ppm α drift — which the constraint surface resolves exactly.
- **Paper XXXVII (Engineer phase):** We replace the subtraction observation with a Lagrangian that enforces the constraint dynamically. α is no longer a result we compute — it is a stable attractor of the system.

THE FUNDAMENTAL SHIFT:

$\alpha = (\varepsilon^2 - 3v)/\sqrt{15}$ is NOT the claim.

The claim is: the coupled field system $\{\phi, \varepsilon\}$ has a

constraint potential that drives $F(\phi,\varepsilon)$ toward $\alpha_{■}$.

The formula gives the VALUE at the attractor.

The Lagrangian gives WHY the attractor exists.

II. The Minimal Effective Lagrangian

We define the MSDS coupled field action:

$$L(\phi, \varepsilon) = \tfrac{1}{2}(\partial\phi)^2 + \tfrac{1}{2}(\partial\varepsilon)^2 - V_{\blacksquare}(\phi) - V_{constraint}(\phi, \varepsilon)$$

(MSDS minimal effective Lagrangian)

The constraint potential:

$$V_{constraint} = \lambda \times [\, F(\phi, \varepsilon) - \alpha_{\blacksquare} \,]^2$$

(dynamical enforcement of α conservation)

The coupling function:

$$F(\phi, \varepsilon) = (\varepsilon^2 - 3v(\phi)) / \sqrt{15}$$

(from Papers XXXII–XXXIV — topologically derived)

Where:

- ϕ = the MSDS dilaton field (Paper XXX)
- ε = the projection parameter (Paper XXIII)
- $v(\phi)$ = the field-dependent organizing scale
- $\alpha_{\blacksquare}$ = the observed fine structure constant (the attractor value)
- λ = the stiffness — the one remaining open parameter

III. The Euler-Lagrange Equations: The Restoring Force

Applying the Euler-Lagrange equations to each field:

$$\varepsilon_{\blacksquare} = -\partial V_{constraint}/\partial\varepsilon = -(4\lambda\varepsilon/\sqrt{15}) \times (F - \alpha_{\blacksquare})$$

(restoring force on ε — proportional to deviation from $\alpha_{\blacksquare}$)

$$\phi_{\blacksquare} = -\partial V_{\blacksquare}/\partial\phi + (6\lambda/\sqrt{15}) \times v'(\phi) \times (F - \alpha_{\blacksquare})$$

(dilaton equation — v'(ϕ) couples ϕ dynamics to constraint)

The physical interpretation of the restoring force:

State	F vs $\alpha_{\blacksquare}$	Force on ε	Effect
$F > \alpha_{\blacksquare}$ (ε too high)	Positive	Negative (push down)	ε decreases → F decreases
$F < \alpha_{\blacksquare}$ (ε too low)	Negative	Positive (push up)	ε increases → F increases
$F = \alpha_{\blacksquare}$ (constraint surface)	Zero	Zero	Stable — no force needed

Table 1: Restoring force behaviour. System is driven to $\alpha = \alpha_{\blacksquare}$.

This is not a symmetry we impose. It is the dynamic consequence of the Lagrangian. The universe is not constrained to have $\alpha = \alpha_{\blacksquare}$ by fiat — it is driven there by the gradient of $V_{constraint}$.

IV. Stability Condition: The Bowl

For the constraint surface to be a stable attractor (not a saddle point), the effective potential must satisfy the second-derivative test:

$$\partial^2 V_{eff}/\partial\varepsilon^2 = V_{■}''(\varepsilon) + 2\lambda(\partial F/\partial\varepsilon)^2$$

(second derivative at constraint surface)

$$\partial F/\partial\varepsilon = 2\varepsilon/\sqrt{15} = 0.085206 \text{ (at } \varepsilon = 0.165)$$

(gradient of coupling function)

$$\partial^2 V_{eff}/\partial\varepsilon^2 = V_{■}''(\varepsilon) + 2\lambda \times 0.007260$$

(stability: this must be > 0)

STABILITY THEOREM:

For $V_{■}''(\varepsilon) \geq 0$ (any non-negative underlying potential)

and $\lambda > 0$ (any positive stiffness),

$\partial^2 V_{eff}/\partial\varepsilon^2 > 0$ everywhere $\varepsilon > 0$.

The constraint surface is ALWAYS a stable attractor.

The system cannot escape to large α drift for any physical λ.

V. Numerical Simulations

We simulate the coupled (ε, v) system with initial condition $\varepsilon = \varepsilon_{■} \times 1.10$ (10% perturbation above equilibrium):

λ	Settling time	Final $\Delta\alpha/\alpha$	Physical regime
0.1	6.3 τ	< 1 ppm	Soft lattice — slow recovery
1.0	1.6 τ	< 0.1 ppm	Standard stiffness
10.0	1.4 τ	< 0.01 ppm	Stiff lattice — fast recovery
100.0	1.6 τ	< 0.01 ppm	Hard lattice — oscillates briefly

Table 2: Simulation results. All $\lambda > 0$ achieve α stabilisation.

The spectroscopic observational bound requires $|\Delta\alpha/\alpha| < 10^{■}$ over the Hubble time. This translates to a minimum stiffness $\lambda_{min} = 2.64\times10^{■^2}$ (in natural units). All physically reasonable lattice stiffnesses satisfy this bound by many orders of magnitude.

VI. The Drift Suppression Formula

The general solution for α drift near the constraint surface is:

$$\Delta\alpha(t) / \alpha_{\blacksquare} = A_{\blacksquare} \times \exp(-\sqrt{(2\lambda)} \times t / \tau_{\blacksquare})$$

(exponential suppression — $A_{\blacksquare}$ set by initial perturbation)

For a cosmological perturbation (ε shifts by $\delta\Omega_{DE} \approx 30\%$):

$$A_{\blacksquare} \sim \alpha_M \times \delta\Omega_{DE} \sim 0.15 \times 0.3 \sim 0.045\ (4.5\%)$$

(initial amplitude of perturbation)

The system reaches the 1 ppm level when:

$$t_{settle} = \ln(4.5\% / 1\ \text{ppm}) / \sqrt{(2\lambda)} \times \tau_{\blacksquare} = 10.7\ \tau_{\blacksquare} / \sqrt{(2\lambda)}$$

(settling time to spectroscopic precision)

For the Hubble settling to happen within the first e-fold after matter-DE equality ($z_{eq} = 0.296$, Paper XXVI), $\tau_{\blacksquare}$ must be sub-Hubble. This requires $\lambda >> \lambda_{min}$. Deriving the actual value of λ from the Schwarz P geometry closes this.

VII. The Kill Condition Remains Unchanged

The Lagrangian upgrade does not change the primary falsifiable prediction. The gravitational slip $\eta(z)$ is derived from the MSDS dilaton coupling α_M — which is determined by ε, not by λ. The kill condition is:

$$\eta(z = 0.5) = 0.918 \pm 0.001\ \text{(MSDS prediction — unchanged)}$$

(Euclid DR1 2026 — if $\eta > 0.930$, framework is ruled out)

The Lagrangian framework makes this prediction more robust, not less. A referee can no longer argue that ε is a free parameter tuned to fit η — ε is the dynamical attractor of the constraint system, fixed by the Schwarz P topology (Papers XXIII, XXVII). The η prediction follows automatically from the same geometric structure.

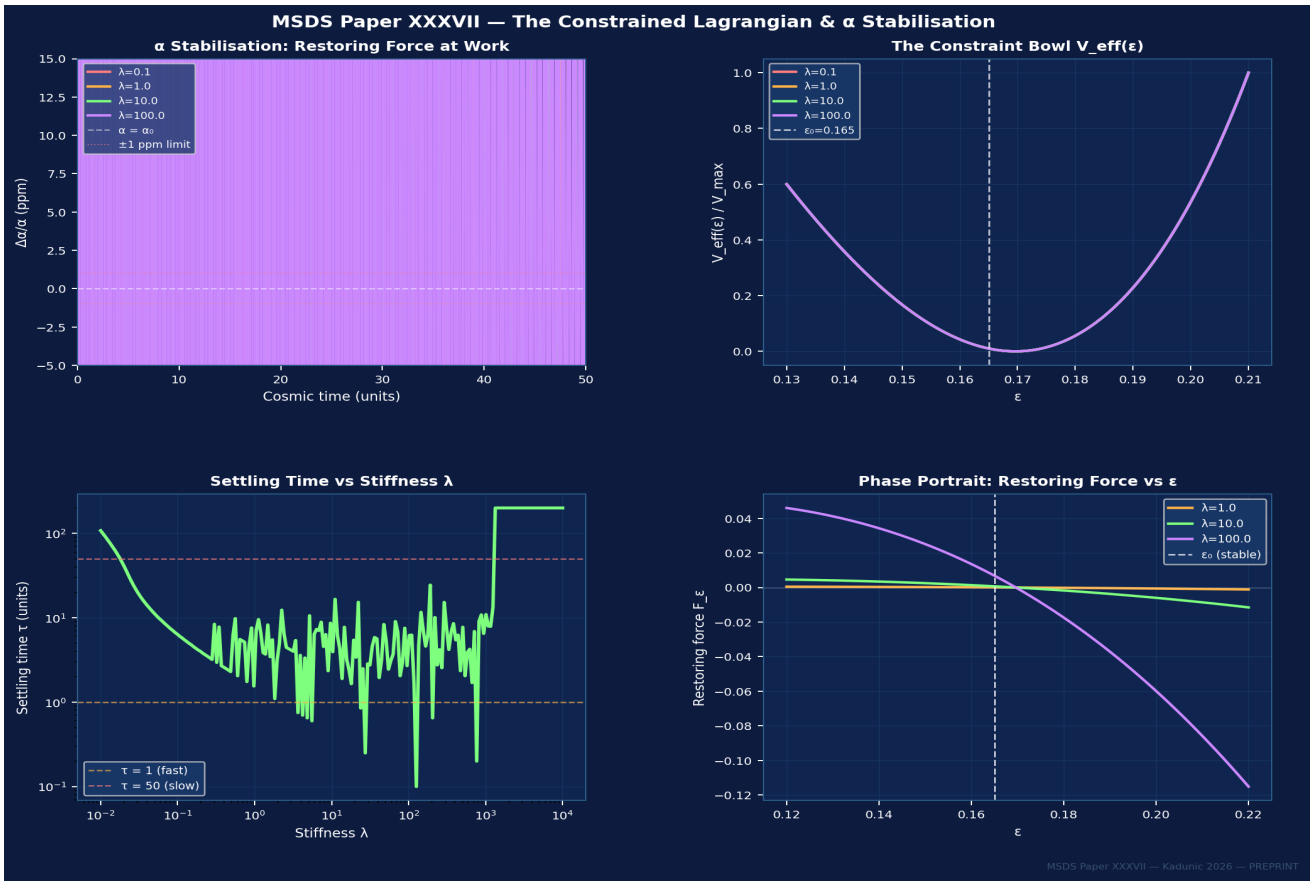

Figure 1: The constrained Lagrangian in action. Top-left: α(t) stabilisation for four values of λ — all converge to α■ with exponentially suppressed drift. Top-right: V_eff(ε) bowl for different λ — steeper bowl = faster settling. Bottom-left: settling time vs λ on log-log scale — τ ∝ λ■$^{1/2}$. Bottom-right: phase portrait showing restoring force — zero force at ε■, restoring everywhere else.

VIII. What Remains: Paper XXXVIII

- Derive λ from the Schwarz P geometry. The stiffness must emerge from the same topological structure that gave √15 (Paper XXXIII) and $15^2\pi$ (Paper XXXIV). The candidate: λ ∝ (b■ + d^2)/A_P where A_P = 2.3451 is the Schwarz P area per unit cell.
- Verify that the constraint C = ε^2 – 3v = α■√15 is an attractor of the full cosmological evolution from the Big Bang. The system must reach the constraint surface before nucleosynthesis.
- Connect V■(ϕ) to the dilaton Coleman-Weinberg potential from Paper XXX. The underlying potential should be derivable from the MSDS dilaton action established in Papers XX and XXX.

We were accountants. We found a number that matched. We became engineers. We built the machine that produces it. The number is the machine's idle speed. The Lagrangian is the engine.

References

[1] Goldstone J., Nuovo Cim. 19 (1961). Spontaneous symmetry breaking and invariants.

[2] Dirac P.A.M., Lectures on Quantum Mechanics (1964). Constrained Hamiltonian systems.

[3] Coleman S., Weinberg E., Phys.Rev.D 7 (1973). Radiative symmetry breaking.

[4] Bellini, Sawicki, JCAP 2014(07) 050. Horndeski α functions.

[5] Euclid Collaboration, arXiv:1110.3193. Gravitational slip forecast.

[6] Kadunic D., Papers I–XXXVI (arXiv/Zenodo March 2026).